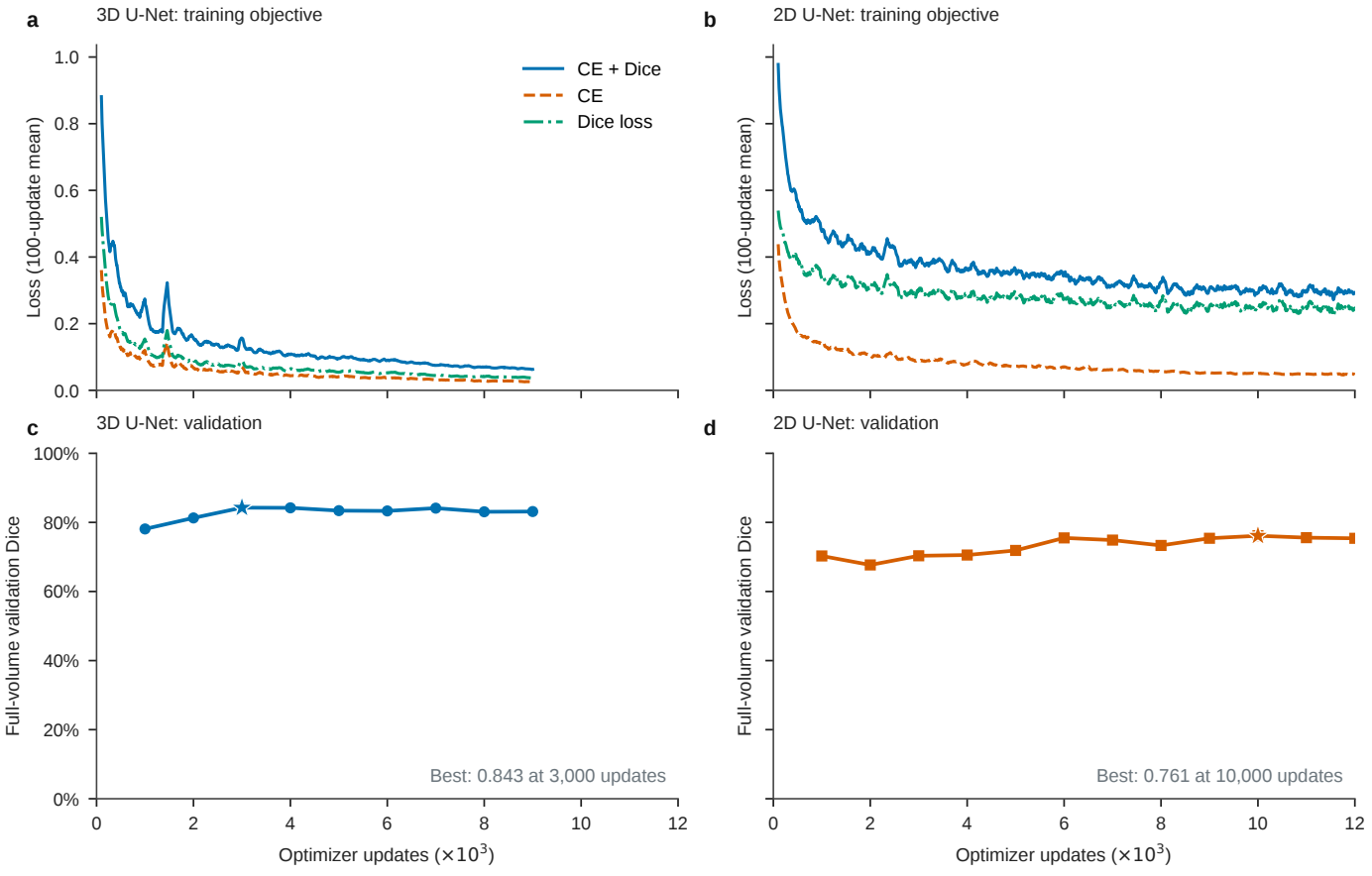

a
3D U-Net: training objective
CE + Dice
CE
Dice loss
Loss (100-update mean)
1.0
0.8
0.6
0.4
0.2
0.0
b
2D U-Net: training objective
Loss (100-update mean)
c
3D U-Net: validation
Full-volume validation Dice
100%
80%
60%
40%
20%
0%
Best: 0.843 at 3,000 updates
0
2
4
6
8
10
12
Optimizer updates (×10^3)
d
2D U-Net: validation
Full-volume validation Dice
Best: 0.761 at 10,000 updates
Optimizer updates (×10^3)